

Graph Theory

Question difficulty is indicated by 🦶 (lower difficulty), 🦶🦶 (greater difficulty), and 🦶🦶🦶 (greatest difficulty). A topic label is also given on the question.

Revision

Problem 1

2015 Kazakhstan National Olympiad 🦶

A full binary tree consists of $n + 1$ levels, where from each vertex two edges go down and one edge comes from above (the top vertex at level 1 has no incoming edge, and vertices at the last $(n + 1)$ -th level have no outgoing edges).

In how many ways can the edges of this tree be coloured in 2^n given colours (each edge is coloured in one colour) so that for each colour, all edges of that colour form a path from some vertex to a vertex on the last level?

Problem 2

2019 Canadian Mathematical Olympiad Problem 5 🦶🦶

A 2-player game is played on $n \geq 3$ points, where no 3 points are collinear. Each move consists of selecting 2 of the points and drawing a new line segment connecting them. The first player to draw a line segment that creates an odd cycle loses. (An odd cycle must have all its vertices among the n points from the start, so the vertices of the cycle cannot be the intersections of the lines drawn.) Find all n such that the player to move first wins.

Problem 3

IMO 2005 🦶🦶🦶

In a mathematical competition, 6 problems were posed to the participants. Every pair of problems was solved by more than $\frac{2}{5}$ of the contestants. Moreover, no contestant solved all six problems. Show that there are at least two contestants who solved exactly five problems each.

Cliques 🦶

Problem 4

Show that $K_{m,n}$ has at most a perfect square number of edges using Turán's theorem, where m, n are even and $m + n$ is constant.

Matchings

Problem 5

There are $2n$ people in a room, where each pair of people is classified as either friends or strangers. Two players play a game where they alternate turns picking a person who has not been picked before and is friends with the person picked on the previous turn.

The last player who can make a legal move wins. The first player may pick any person. Prove that the second player has a winning strategy if and only if the $2n$ people can be partitioned into n disjoint pairs such that the two people in each pair are friends.

Problem 6

USA TSTST 2018 Problem 2 In the nation of Onewaynia, certain pairs of cities are connected by one-way roads. Every road connects exactly two cities (roads are allowed to cross each other, e.g., via bridges), and each pair of cities has at most one road between them. Moreover, every city has exactly two roads leaving it and exactly two roads entering it.

We wish to close half the roads of Onewaynia in such a way that every city has exactly one road leaving it and exactly one road entering it. Show that the number of ways to do so is a power of 2 greater than 1 (i.e. of the form 2^n for some integer $n \geq 1$).